LECTURE 7

Def'n A series $\sum_{n=1}^{\infty} a_n$ is called an <u>alternating series</u> if a_n is an alternating sequence.

Theorem <u>Alternating Series Test</u>

For an alternating series $\sum_{n=1}^{\infty} a_n$ with $|a_n| = b_n$, then if:

(1) $\lim_{n \to \infty} b_n = 0$, and

(2) b_n is decreasing, then $\sum_{n=1}^{\infty} a_n$ converges.

Ex Determine if the series converges or diverges.

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}} \qquad \text{Let } b_n = \frac{1}{\sqrt{n}}$$

1) $\lim_{n \to \infty} b_n = \lim_{n \to \infty} \frac{1}{\sqrt{n}} = 0$

2) $b_n = \frac{1}{\sqrt{n}} > \frac{1}{\sqrt{n+1}} = b_{n+1}$

Thus, $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$ converges by the AST.

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1} n}{e^n} \qquad \text{Let } b_n = \frac{n}{e^n}.$$

1) $\lim_{n \to \infty} b_n = \lim_{n \to \infty} \frac{n}{e^n} \overset{H}{=} \lim_{n \to \infty} \frac{1}{e^n} = 0$

alternating series, alternating series test

CONTINUED

2) Let $f(x) = \frac{x}{e^x}$. Then $f'(x) = \frac{e^x - xe^x}{e^{x^2}} < 0$.

when $x > 1$.

Thus, $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} n}{e^n}$ converges by the AST.

Theorem If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.

Note The converse is false.

Def'n For a series $\sum_{n=1}^{\infty} a_n$, if $\sum_{n=1}^{\infty} |a_n|$ converges,

then $\sum_{n=1}^{\infty} a_n$ is called <u>absolutely convergence</u>.

If $\sum_{n=1}^{\infty} a_n$ is not absolutely convergent, but

converges, then it is <u>conditionally convergent</u>.

Ex Determine if $\sum_{n=1}^{\infty} \frac{(-1)^n \cdot 2n}{n^3+1}$ is absolutely

convergent, conditionally convergent, or divergent.

Consider $\sum_{n=1}^{\infty} \left| \frac{(-1)^n \cdot 2n}{n^3+1} \right|$ we have $\sum_{n=1}^{\infty} \frac{2n}{n^3+1}$.

Let $a_n = \frac{2n}{n^3+1}$, $b_n = \frac{n}{n^3} = \frac{1}{n^2}$. Then,

$$\lim_{n \to \infty} \frac{a_n}{b_n} = \lim_{n \to \infty} \frac{2n}{n^3+1} \cdot \frac{n^2}{1} = \lim_{n \to \infty} \frac{2}{1 + \frac{1}{n^3}} = 2$$

Since $0 < L < \infty$, and $\sum_{n=1}^{\infty} b_n = \frac{1}{n^2}$ converges

by PST, by LCT, the series converges. Hence it is absolutely convergent.

absolutely convergent, conditionally convergent